Experiment No.:4

Name of Experiment: Implement basic R operation (data input,missing values, importingdata into R using different formats: xlsx, CSV, Text files)

SOURCE CODE:

1. Data Input in R

a. Manual Data Input

You can create small datasets directly in R using vectors or data frames.

```
Creating vectors
names <- c("John", "Alice", "Bob")
age <- c(23, 25, NA)   NA is used for missing values
gender <- c("M", "F", "M")

Creating a data frame
df <- data.frame(Name = names, Age = age, Gender = gender)

View the data frame
print(df)
```

2. Handling Missing Values

Missing data in R is represented by NA. Here are some ways to handle them:

a. Identify Missing Values

You can check for missing values using is.na().

```
Check if values are missing
is.na(df)

Count the number of missing values in each column
colSums(is.na(df))
```

b. Remove Missing Values

Remove rows with missing values using na.omit().

```
Remove rows with missing values
df_clean <- na.omit(df)

View cleaned data
print(df_clean)
```

c. Impute Missing Values

You can replace missing values using functions like mean() or median().

```
Replace missing Age values with the mean of the non-missing values
df$Age[is.na(df$Age)] <- mean(df$Age, na.rm = TRUE)

View the updated data
print(df)
```

3. Importing Data into R

R allows you to import data from various formats such as CSV, Excel (XLSX), and text files.

a. Importing a CSV File

The read.csv() function is used to import CSV files.

```
Import CSV file (make sure the file path is correct)
data_csv <- read.csv("path_to_file.csv")

View the first few rows
head(data_csv)
```

b. Importing an Excel File (XLSX)

To import Excel files, you need to install and load the readxl package.

```
Install the readxl package if you haven't already
install.packages("readxl")

Load the readxl library
library(readxl)

Import the Excel file (specify the sheet if necessary)
data_xlsx <- read_excel("path_to_file.xlsx", sheet = 1)

View the first few rows
head(data_xlsx)
```

c. Importing a Text File

You can use read.table() or read.delim() to import text files.

```
Import a text file with tab-separated values
data_txt <- read.table("path_to_file.txt", header = TRUE, sep = "\t")

View the first few rows
head(data_txt)
```

d. Importing Data from a Delimited File

You can use read.csv() or read.delim() to load data from different delimited formats.

```
 Import a comma-separated file
data_csv2 <- read.csv("path_to_file.csv", sep = ",")

 Import a tab-separated file
data_tab <- read.delim("path_to_file.txt", sep = "\t")

 View the data
head(data_tab)
```

Use Cases:

- Manual Data Input: Use data.frame() to create datasets manually.
- Handling Missing Values: Use is.na() to identify, na.omit() to remove, and functions like mean() to impute missing values.
- Importing Data:
    - CSV: read.csv()
    - Excel: read_excel() from the readxl package
    - Text: read.table() or read.delim()

These basic operations allow you to input and clean data before performing any analysis.

Basic Level Questions

1. Which function is used to create a data frame in R?
   Answer: data.frame()
2. What does NA represent in R?
   Answer: Missing values
3. Which function is used to import CSV files in R?
   Answer: read.csv()
4. How do you check for missing values in a dataset?
   Answer: is.na()
5. Which R package is used to read Excel files?
   Answer: readxl

Moderate Level Questions

1. Write R code to remove rows with missing values from a data frame.
   Answer: df_clean <- na.omit(df)
2. How can you count the number of missing values in each column?
   Answer: colSums(is.na(df))
3. Which function is used to read a tab-separated text file?
   Answer: read.delim()
4. How would you view the first few rows of an imported dataset?
   Answer: head(data)
5. Write R code to impute missing Age values with the mean of the column.
   Answer:

   df$Age[is.na(df$Age)] <- mean(df$Age, na.rm = TRUE)

Advanced Level Questions

1. How can you import a specific sheet from an Excel file in R?
   Answer:

   data <- read_excel("file.xlsx", sheet = 2)

2. Explain how to handle different delimiters when reading a file.
   Answer: Use sep parameter in read.csv() or read.table() (e.g., sep = "," or sep = "\t")
3. Create a complete workflow that inputs data, cleans missing values, and displays the cleaned data.
   Answer:

```
data <- read.csv("file.csv")
data$Age[is.na(data$Age)] <- mean(data$Age, na.rm = TRUE)
print(data)
```

4. What is the difference between **read.table()** and **read.delim()**?
   Answer: read.delim() is a wrapper for read.table() with default separator as tab (\t).
5. How do you install and load a package only if it is not already installed?
   Answer:

   ```
   if(!require(readxl)) install.packages("readxl")
   library(readxl)
   ```

Use Cases:

1. Manual Data Entry Using data.frame()
2. Detecting and Counting Missing Values
3. Removing Rows with Missing Data
4. Imputing Missing Data Using Mean or Median
5. Importing Data from CSV, Excel, and Text Files